- The acceleration due to gravity is denoted by $g\,\mathrm{m\,s^{-2}}$. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.

12 In this question x and y are the horizontal and upwards vertical directions respectively.

An astronaut is standing on the surface of the moon exploring the motion of a ball.

(a) The astronaut drops a ball from rest from 1 m above the surface. It takes 1.1 s to hit the surface.

Calculate the value of the acceleration due to the moon's gravity. Give your answer correct to **3** significant figures. **[2]**

The astronaut stands in a crater of the moon and hits the ball with a golf club from the moon's surface. The initial velocity of the ball is $25\,\mathrm{m\,s^{-1}}$ at an angle of 40° above the horizontal in the x-direction.

(b) Taking the origin to be the point of projection, determine the equation of the trajectory of the ball. Give your answer in the form $y = \mathrm{f}(x)$, with each of the coefficients correct to **3** significant figures. **[4]**

(c) The edge of the crater is 40 m away from the point of projection and 15 m above it.

Determine whether the ball goes over the crater's edge. **[2]**